



Oil palm fresh fruit bunches (FFB) growth determination system to support harvesting operation

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Abstract

Oil palm fresh fruit bunches (FFB) need to be harvested at the optimum maturity stage to optimise the quality of palm oil. Currently the oil palm harvester manually determines the FFB maturity based natural indicators such as FFB color appearance and number of FFB loose fruit drops under the tree. During executing the harvesting operation the harvester need to search for a ripe FFB and at the same time carrying the harvesting pole. Tedious harvesting operation degrades the consistency of their judgment. The harvested FFB must be screened at the oil palm mill to separate into groups of maturity level according to the standard. A computer application called Maturity Table was developed to manage the FFB maturity information in form of digital images. The development of FFB from the anthesis to harvesting stage was monitored by using a handy digital camera. The digital images of the FFB were processed by using digital image processing techniques to extract the color that represent the maturity stages. As a result from the analysed image information and tabulated data in Maturity Table, a relationship of FFB color changing and maturity stages were investigated. The processed information also enables the development of a map displaying the location of FFB and the maturity stages. The developed map can be used as a support system in harvesting operation. This support system enables site specific harvesting at optimum maturity stage, overcome losses due uncollected loose fruit, potential application to eliminate FFB screening process at oil palm mill level and potential tool to predict FFB yield as well as plantation resources management.

Key words: Harvesting support system, site specific harvesting, Maturity Table, oil palm yield.

Introduction

Oil palm (*Elaeis guineensis* Jacq var. Tenera) is one of the most important plantation crops in Malaysia. Oil palm fresh fruit bunch (FFB) harvesting and collection form a single largest direct cost in the production of oil palm ^{1,2}. FFB yield is a common parameter to measure the productivity and mostly result verified and confirmed at post harvesting level. Harvesting oil palm FFB at right stage of ripeness is critical to ensuring optimum quality and quantity of oil production and thus profitability of the industry ³.

In the standard operating procedure, matured FFB is highly related with numbers of loose fruits drops under the oil palm tree. This is used as a sign to harvest the FFB and has a drawback because uncollected loose fruits are one of the factors contributed to the losses in Malaysian oil palm industries. Sime Darby Plantation, one of the biggest oil palm plantation in Malaysia, reported for every uncollected loose fruit, the company will potentially lose almost RM11 million in annually income ⁵. A new kind of monitoring system is required to determine maturity stages of oil palm to avoid losses due to loose fruit ⁶.

Various techniques have been developed to measure the maturity stages of FFB. Two parameters commonly used to measure the maturity of FFB are color, fruitlets moisture and oil content. FFB maturity stage and location are very important information. Farmer or plantation manager can use this information to manage their workers, machines and other agricultural resources

to increase productivity. This information is also useful for yield forecasting and crop management.

Each oil palm tree potentially produced 10-12 FFB/tree annually ⁶. Female inflorescence takes between 4.5 to 6 months to develop from anthesis to ripeness. The synthesis of the oil in mesocarp occurs from 15 to 16 weeks after anthesis and completed in about 20 to 22 weeks. In order to maximise the oil extraction rate, the FFB should be harvested at its peak of ripeness ⁷. The harvesting frequency was scheduled within 10-12 days interval. In every cycle of harvesting operation, farmer does not have any information on how many bunches and which oil palm trees will be harvested.

During executing the harvesting operation the harvester will carry a sickle attached to bamboo or aluminium pole for palm tree over 12 m height and chisel for shorter palm. They will move to each tree searching for ripe fruit (including tree without ripe FFB) and used their own experience to guess the maturity stage based on the natural indicators. Tedious harvesting task force them to lose their judgment consistency and concentration. Their task will be easy if they know the exact location of ripe FFB. A reliable FFB maturity stage determination system and its specific location will be very useful towards better oil palm crop management system. Farmer can schedule and execute targeted harvesting effectively if they know the location of ripe FFB. Worker salary per harvested bunch and total count of bunches can be easily

projected before completing the task and yield for every harvesting cycle easily justified after completing the harvesting operation.

Khalid and Abbas⁸ and Abbas *et al.*⁹ utilize the FFB mesocarp moisture and oil content relationship to develop a microwave based sensor to measure the maturity stages. This sensor only suit for laboratory condition. Razali *et al.*^{10, 11} and Wan Ishak *et al.*¹² developed a model to determine the FFB maturity, hue value of the FFB color image was correlated with the FFB oil content and the coefficient of determination is $R^2 = 0.884$. This will help farmer to confirm the maturity stage of the FFB at field during executing the harvesting operation.

Wan Ishak *et al.*¹³ and Alfatni *et al.*¹⁴ used RGB color camera to capture the image of harvested FFB. The intensity values of R (red), G (green) and B (blue) were analyzed to differentiate the FFB into categories namely black, hard, ripe, over ripe, empty and rotten. The experiment was conducted under controlled environment in the lab condition. Result has shown that average value of red color has highest difference value among other colors in the categories. Wan Ishak *et al.*¹³ suggested to extent the capability of the developed vision system to deal with real plantation environment.

According to Arokiasamy¹⁵, color of matured FFB was not a constant feature and varies between individual palms and between geographic areas. To have an accurate estimation of FFB ripeness based on RGB color index required more than 100 samples per class, and the percentage of accuracy was around 90 percent¹⁶. Junkwon *et al.*¹⁷ developed a system to estimate the weight and ripeness of FFB using color and hyper spectral imaging. The RGB color images were transformed into HSI (hue, saturation and intensity) color space. In hyper spectral imaging, FFB image was acquired within the wavelength of 400 - 1000 nm. Junkwon *et al.*¹⁷ concluded that ripeness estimation using hyper spectral imaging achieved acceptable accuracy and possible for a practical use to estimate FFB ripeness, but the cost for the equipment is not affordable, even for bigger plantation. As stated by Osborne *et al.*⁵, a new kind of maturity determination is required that is simple and affordable.

Accurate yield estimation will be very useful to plan labor and machinery requirement, monetary budget, oil palm mill capacity and various oil palm plantation management aspect. Ghani *et al.*⁶ described a method to estimate the FFB yield, termed as "Black Bunch Count". Black bunch count is a process of counting a bunch of age 1.5 months after anthesis including ripe and unripe bunch at a selected oil palm tree. Recommended amount of tree sample is 5 to 10 percent from the total tree in a block of 0.3 to 0.5 km² area. Estimation of FFB yield is a result of multiplying the total counted bunches with the estimated bunch weight.

This practice motivates us to develop a software based system to support this task. The objectives of this research were to develop a FFB growth monitoring system to determine the harvesting date of particular FFB at particular oil palm tree and the projection of the expected FFB yield of the particular oil palm tree for the whole of its economic life.

Methodology

Image acquisition: This research was carried out at the Universiti Putra Malaysia oil palm field at the coordinate of 2°58'47.97"N and 101°43'44.77"E. Sixty oil palm trees with the age of five years were selected to be monitored. In this research a fruiting cycle of oil palm FFBs was observed within the period of fruit development starting from the early FFB development until the day of

harvesting. Oil palm FFB takes 17 to 24 weeks to ripen from the anthesis stage to maximum oil content synthesis. According to Corley and Tinker¹⁸, anthesis of normal FFB inflorescence starts when the trilobes flower open and after anthesis the trilobes flower change to purplish color. Anthesis in normal inflorescence usually lasts for 36-48 hour but may extend to a week. Based on this information image acquisition was carried out in three days interval to acquire the images of FFB spanning from the inflorescence stage to harvesting stage.

The process started in the middle of May 2010 extended to 8 months until end of December 2010. Panasonic DMC-TZ10 camera with embedded global positioning system (GPS) receiver was used to acquire the FFB images. Each oil palm trees, inflorescences and FFBs under study were tagged and given a unique identity for data recording and retrieval purposes. The images of FFB were acquired independently without specific camera or FFB orientation and time of acquisition. By the end of December 2010, a total 809 FFB at various maturity stages were monitored, resulted at nearly 30000 FFB images.

Image data management: Data management is a very important activity in precision agricultural practice. A systematic data management was developed. Each image was named according to the given tag and stored in computer using JPEG* digital format. A tabulated form of image information, so called Maturity Table, was developed to ease the data management.

FFB images grouping based on maturity stages: A complete FFB development from anthesis to harvesting was named as 'Full Set FFB'. The images contain a physical appearance of FFB from early development until the days of harvesting. Information from the Full Set FFB can be utilised to determine the maturity stages of the FFB in the acquired images. Determination of maturity stages was measured from anthesis stage to harvesting stage. The date of image acquisition of opened trilobes flower was recorded as the date of maturity stage in week 0 to indicate the beginning of the FFB development. The last image in the Full Set FFB images will be the image of FFB before harvesting. It can be on the same day of harvest or within three days before harvesting (three days interval image acquisition).

A simple equation to calculate harvesting date was embedded in the each cell of the Maturity Table. An estimated age of the FFB during first image being captured was recorded in the cell accordingly with the date of acquired image. The equation calculates the maturity stages of the FFB in week, the result automatically appeared in Maturity Table. To rectify the accuracy of the estimated age, the processed images data from Full Set FFB were used because in the Full Set FFB the date of anthesis and harvested bunches were known.

Fig. 1(a) is the image of FFB 12E, one of the Full Set FFB. It was at the beginning of anthesis stage (week 0) on 15 May 2010. A zero week was inserted in the cell of the Maturity Table and it automatically added the number. Termination of the addition was on the date of the last image before harvesting which was on 6 October 2010. Maturity Table determined the FFB 12E was harvested on 9 October 2010, according to the value in the table it was in the first day in week 21.

Each maturity stages have an interval of seven days. The maturity table calculates the maturity stages of the Full Set FFB based on



*Joint Photographic Expert Group. The name of the committee that created the JPEG standard

date of the images in week 0. In order to group the Full Set FFB images in their respective maturity stages, the dates of image acquisition were matched accordingly with the date and respective weeks in the Maturity Table.

Non-full set FFB: In the monitoring process not all FFBs were monitored at their early development from anthesis. This set of images was termed as Non – Full Set FFB. There were two types of Non-Full Set FFB, Type A and Type B. Non-Full Set FFB type A was the set of FFB in which the images were acquired after anthesis stage to harvesting stage. FFB images acquired before anthesis stage but not complete to harvesting stage were categorized as Non-Full Set FFB type B. The difference between type A and B was their maturity stage determination approach. Maturity stages for type A need to be determined by the harvester and then rectified by using maturity table. While maturity stages type B can be determined

directly from Maturity Table since the anthesis stage is known.

The maturity stages of the non-Full Set FFB were determined by visual comparison of the color appearance in the images between non-Full Set FFB type A and Full Set FFB of particular oil palm tree. A similar color appearance of the Non-Full Set FFB type A with Full Set FFB from the same tree was assumed to be within the same week of maturity stage. Based on this assumption the estimated age of Non Full Set FFB type A was rectified using the Maturity Table. For example, Fig. 2 shows a Non-Full set FFB Type A for FFB 14A (34*) dated 18 September 2010. This bunch was harvested on 21 September 2010 and no image was available on that date.

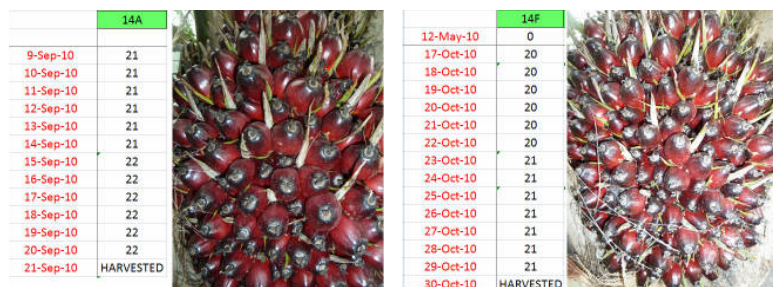
was available on that date.

Full set FFB 14F was chosen as an image comparator. Images from FFB 14A were visually compared with images from FFB 14F. This process only considers similar color appearance. It was found that image 14A (32) dated 9 September 2010 has a similar color appearance with image 14F (37) dated 27 October 2010. By referring to the Maturity Table, FFB 14F on the 27 October 2010 have a maturity stage of week 21. Then in column 14A at the date of 9 September 2010 was inserted with number 21. It was found that on 21 September 2010 FFB 14A was probably harvested at maturity stage of week 22

For verification purposes both images of 14A (32) and 14F (37) were processed to extract the hue values. Kmeans clustering



a



b

c

*nth number of images from a particular FFB.

Figure 2. Example of maturity determination for Non-Full Set FFB Type A. a) FFB 14A (34), on 18 September 2010. b) FFB 14A(32) on 9 September 2010. c) FFB 14F (37) on 27 October 2010.

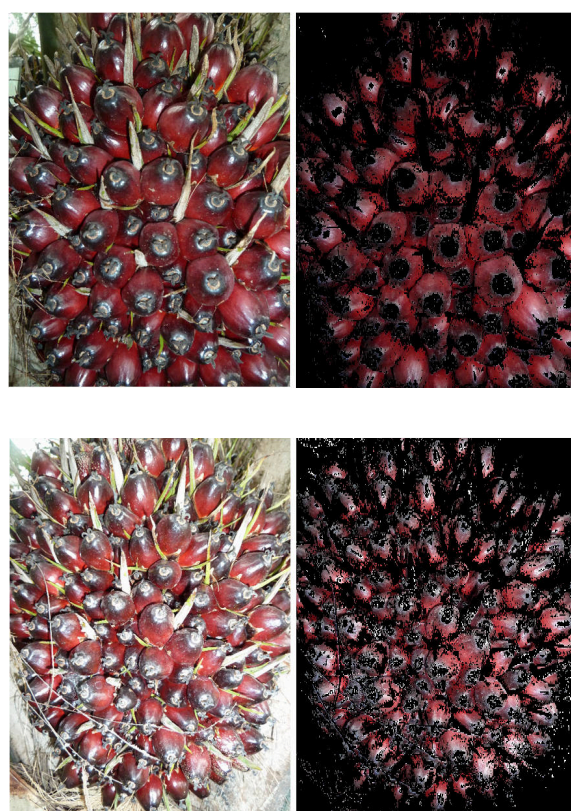


Figure 3. Processed image by using Kmeans clustering technique: a) and c) the original image of 14A(32) and 14F(37) respectively, b) and d) the image of 14A(32) and 14F(37) after K means clustering process.

Table 1. Hue value of the processed images to rectify maturity stage.

14F				14A			14A	
FFB images	Date	Week	Hue ^a	FFB images	Date	Estimated Week	Hue	Rectified Week
14F(34)	18-Oct	20	0.776	14A(29)	25-Aug	19	0.892	21
14F(35)	21-Oct	20	0.837	14A(30)	28-Aug	19	0.913	21
14F(36)	24-Oct	21	0.82	14A(31)	31-Aug	19	0.92	21
14F(37)	27-Oct	21	0.923	14A(32)	9-Sep	21	0.965	23

^a Numerical value representing color of a pixel (picture element) in a digital image.

technique was used to extract the desired image of the FFB as shown in Fig. 3.

This procedure was continued by processing all the images in the maturity stage at week of 21 for both FFB 14A and FFB 14F as shown in Table 1. The hue values of FFB 14F in week 21 were in the range of 0.82 to 0.92. The maturity stage expected will have a similar color appearance by applying the same assumption as manual judgement. Maturity stages of 14A were rectified by adding two more weeks based on similar range of hue value of 14F. Previously the image of FFB 14A (32) was categorised as week 21 and was found to be in week 23 after rectification process using image processing technique.

Results and Discussion

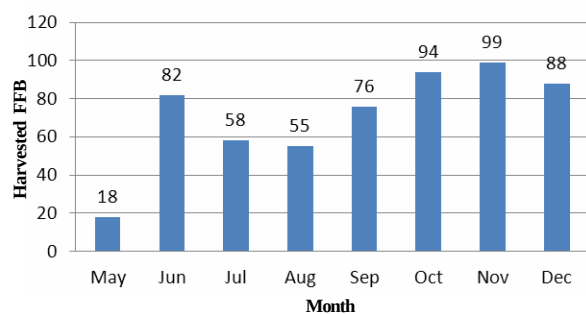
Maturity Table: The developed Maturity Table was a very useful tool in recording the data. It was very helpful in the process of sorting and grouping the digital images in their respective maturity stages. Table 2 shows the recorded data. A total of 809 FFB were monitored, 217(24%) of Full Set FFB, 351(44%) of Non Full Set Type A, 237(31%) Non Full Set Type B and 4(1%) of undeveloped FFB. These data can easily be transformed into graphs, charts or tables for further interpretation.

Table 2. Fraction of monitored FFB.

FFB Type ^a	FFB (Bunches)	Percentage (%)
Full Set	217	24
Non Full Set Type A	351	44
Non Full Set Type B	237	31
Undeveloped FFB	4	1
Total FFB	809	100

^a Full Set-FFB data collected from anthesis to harvesting. Non Full Set Type A- FFB data collected after anthesis to harvesting. Non-full set Type B- FFB data collected from anthesis to stages before harvesting.

Harvesting data: From the recorded data, 217 Full Set FFB and 351 Non Full Set Type A FFB were harvested from May to December 2010. Fig. 4 shows the bar chart of the harvested FFB from May to December 2010. The bar chart shows the fluctuation in the recorded yield from May to December 2010. This fluctuation pattern might have relation with external uncontrolled parameter such as weather. This is not the scope of study of this research,

**Figure 4.** Harvested FFB during May to December 2010.

but from this result we can observe the occurrence of this pattern with the help of well recorded monthly yield data extracted from Maturity Table.

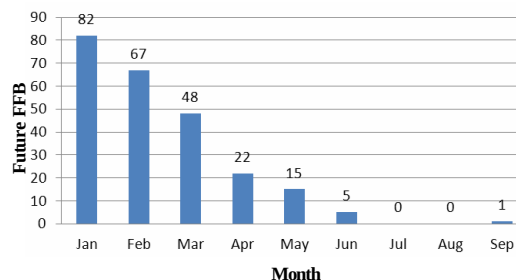
On December 2010 there were 237 Non Full Set FFB Type B. These Non Full Set FFB Type B were expected to be harvested within January to September 2011 as projected using

Maturity Table. The harvesting dates of these Non Full Set FFB Type B FFB were also generated by using the Maturity Table. Fig. 5 shows the future FFB yield from January to September 2011. Future FFB decreased from 82 bunches in January 2011 to 5 bunches in June 2011. This data shows that Maturity Table can produce a projection of FFB yield of six months ahead. Theoretically the projection for this maturity was reasonable. The decreasing pattern in future FFB as in Fig. 5 was a reliable data and this might happened because of the influences of external factors. As a comparison, the yield of 82 bunches in June 2010 and 5 bunches in June 2011 is obviously a sign of improper field management.

A digital map plotting the variability of FFB yield can be produced using any GIS software. In the form of digital map the user can visualize the yield variability and easily understand the graphic representation. Fig. 6 shows the yield map of the oil palm in this research during May 2010 to September 2011. It shows a yield decreasing pattern from North to South of the research field. The user can also conclude the yield variability due to difference of soil properties such as nutrient and moisture content. Based on the information from a digital map proper action can be carried out to rectify the yield variability.

Harvesting schedule: Since FFB were harvested at different optimum maturity stage, it is important to determine best optimum maturity stage of each oil palm tree. User can choose the best optimum maturity stage based on the highest oil content and respective FFB color appearance. In this study the mean of the harvested FFB age of Full Set FFB was taken as the best optimum maturity stage. This approach considers the judgment of harvester during selecting the best FFB to harvest. Table 3 shows the mean of the optimum maturity stage of oil palm trees in this study.

This mean value was used by the Maturity Table to calculate the optimum maturity stage of each oil palm tree and generate the harvesting date to harvest the future FFB. For example oil palm tree P1 has an optimum maturity stage at week 21. Maturity Table will calculate the FFB development in tree P1 from the opened trilobes flower and terminate the calculation in week 21. The date

**Figure 5.** Expected FFB harvested during January to September 2011.

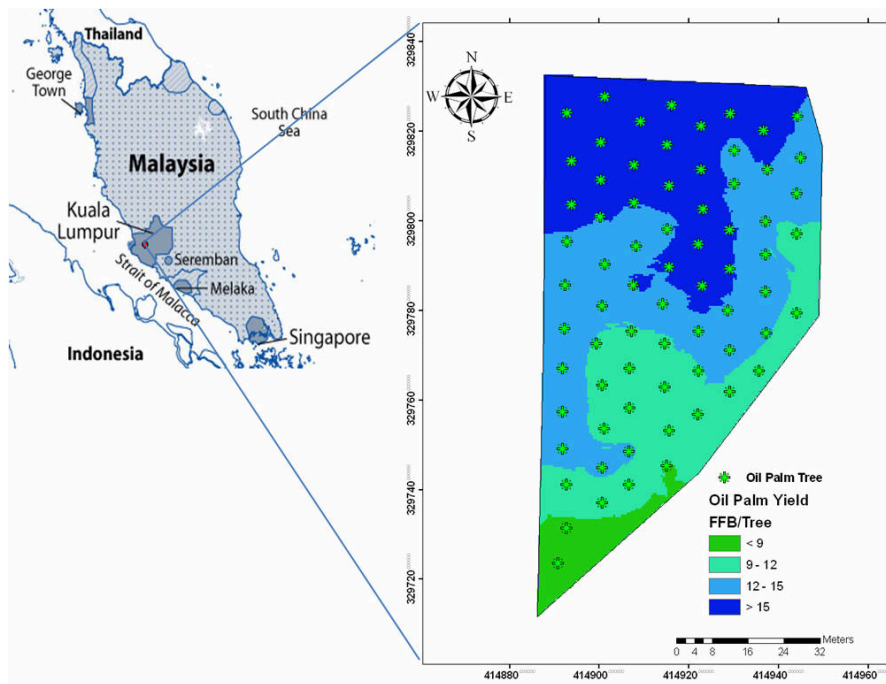


Figure 6. Map of yield variability of the research area.

Table 3. Mean harvesting age of optimum maturity stage of each oil palm in the study area .

TREE	P1	P3	P6	P7	P9	P10	P11	P12	P13	P14	P15	P16	P17	P18	P19
WEEK	21	23	22	23	23	22	21	21	21	20	23	22	23	20	24
TREE	P20	P22	P23	P24	P25	P26	P27	P28	P31	P32	P33	P35	P36	P37	P38
WEEK	24	23	22	22	21	22	21	22	21	20	23	21	22	23	21
TREE	P39	P40	P41	P42	P44	P45	P46	P47	P48	P49	P50	P51	P52	P53	P54
WEEK	25	24	23	22	21	21	24	21	21	20	22	22	23	24	20
TREE	P55	P56	P57	P58	P59	P60	P61	P62	P65	P66	P68	P70	P71	P72	
WEEK	18	21	22	25	25	22	22	23	21	24	22	23	21	21	

of harvesting is the date that matches according to week 21. A harvesting schedule can be generated from this information.

Harvesting map: Based on information from Maturity Table a harvesting schedule can be tabulated. For example in May 2011 there were 15 FFB expected to be harvested. Table 4 shows the harvesting date and the GPS location of the particular tree, also the identity of the tagged FFB.

Table 4. Harvesting schedule for May 2011.

HARVESTED DATE	FFB	EASTING ^a	NORTHING ^b
3-May-11	16M	414930.1	329808.2
3-May-11	20L	414929.1	329771.2
7-May-11	19J	414928.6	329780.0
10-May-11	59S	414900.6	329744.9
11-May-11	17P	414929.1	329798.1
14-May-11	16L	414930.1	329808.2
17-May-11	62F	414892.7	329731.5
18-May-11	60I	414900.6	329737.0
24-May-11	22O	414922.1	329756.9
24-May-11	62G	414892.7	329731.5
25-May-11	19K	414928.6	329780.0
25-May-11	20I	414929.1	329771.2
25-May-11	40K	414906.7	329741.2
29-May-11	20K	414929.1	329771.2
31-May-11	59R	414900.6	329744.9

Grid map reference system. ^a The grid numbers on the east-west (horizontal) axis. ^b The grid numbers on the north-south (vertical) axis.

The GPS location of each oil palm tree helps user to find the specific location of the FFB to speed up the harvesting operation. A map of harvesting point can be generated based on the information generated in the harvesting schedule. User can plan the harvesting route efficiently. Fig. 7 shows the generated harvesting map based on the information of projected yield in May 2011.

Conclusions

This research produces a simple solution in form of data recording and management system called Maturity Table. By using Maturity Table the digital images were categorized into 4 groups namely Full Set FFB, Non-Full Set FFB Type A, Non-Full Set FFB Type B and Undeveloped FFB. The images in these groups will be further processed using image processing technique to determine the maturity stages. This process will not be reported in this paper, but the output from the processed images must be verified with the result produced by Maturity Table to correctly determine the maturity stages. The data shows that the FFB development from anthesis to ripeness can be calculated by using Maturity Table. FFB yield can be projected monthly and yield per tree can also be projected. By using this information user can produce the harvesting table consisting of the specific location of the mature FFB and the number of FFB available to be harvested, this helps them to manage the harvesting operation properly. The harvester

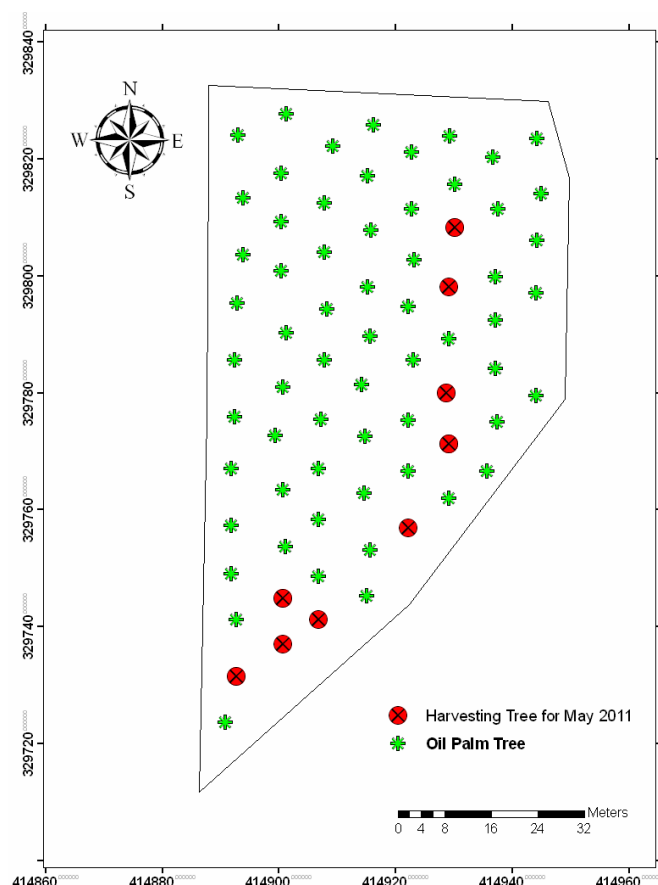


Figure 7. Generated harvesting map for May 2011.

can be allocated efficiently, thus time efficient harvesting operation since the location of mature FFB is known.

The reliability of this support system depends on the availability of the data. The data must constantly deliver to the computer support system. Data collection is a continuous operation along the economic lifetime of the tree. Since the development to ripeness took about six months, this developed system must have a data within a cycle of six month. Application to bigger scale plantation will increase this burden. An automated system for data collection process needs to be develop in the form of robotized equipment carrying the tools to collect the data for bigger size plantation.

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